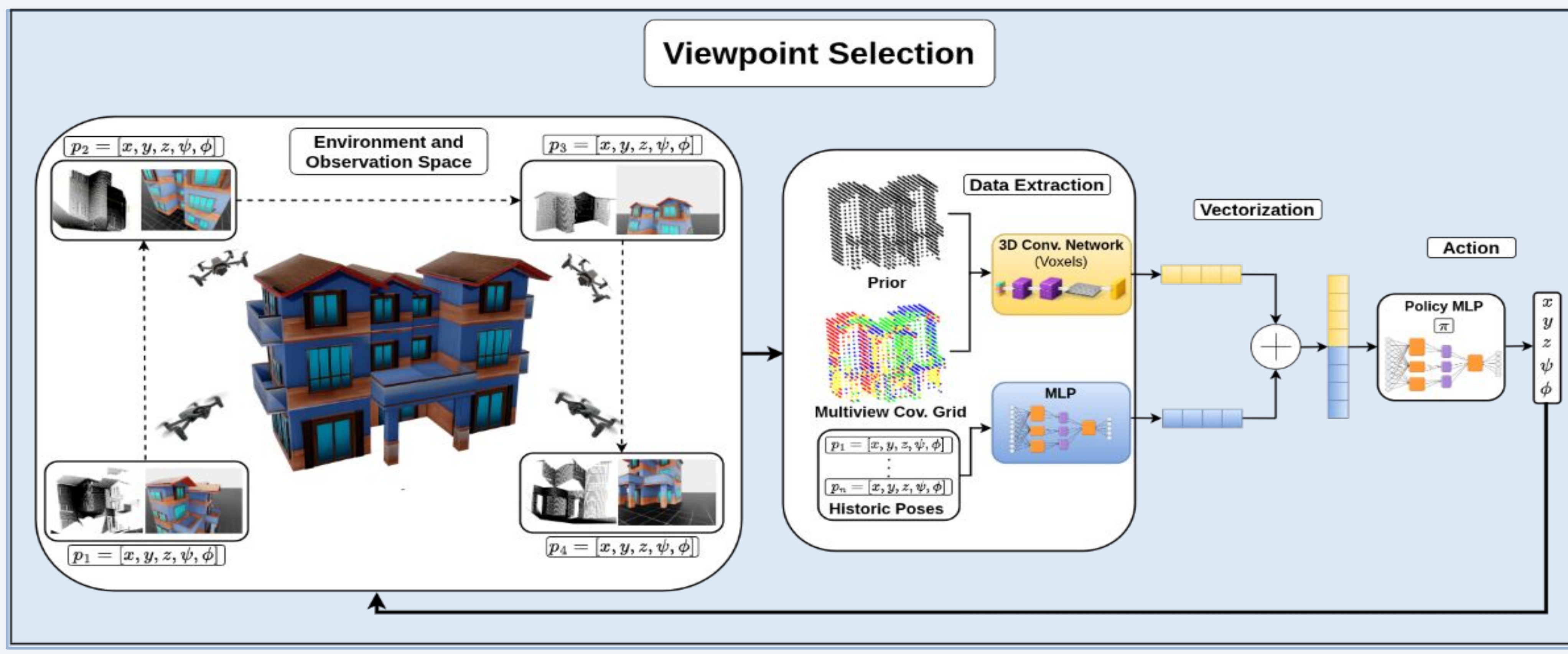
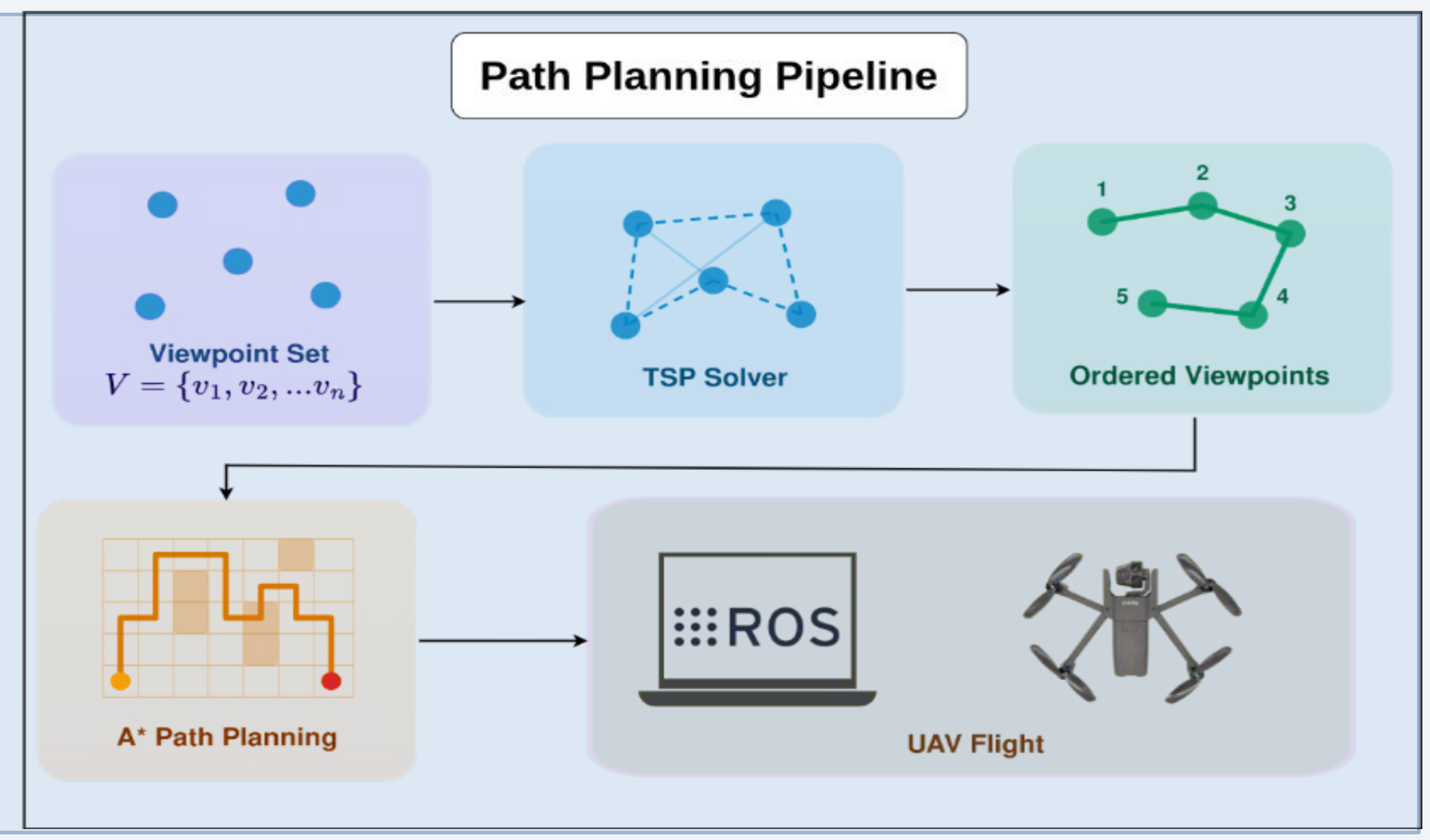
# Autonomous Multi-View UAV Inspection via Reinforcement Learning with Sim-to-Real Validation

**Paulo V. G. Simplicio, Guilherme A. S. Pereira**

Field and Aerial Robotics Laboratory, Department of Mechanical, Materials, and Aerospace Engineering,
Benjamin M. Statler College of Engineering and Mineral Resources

**A complete RL-based framework to autonomously inspect industrial structures. From Sim-to-Real with zero hand-crafted rules and zero structure-specific tuning**




## INTRODUCTION

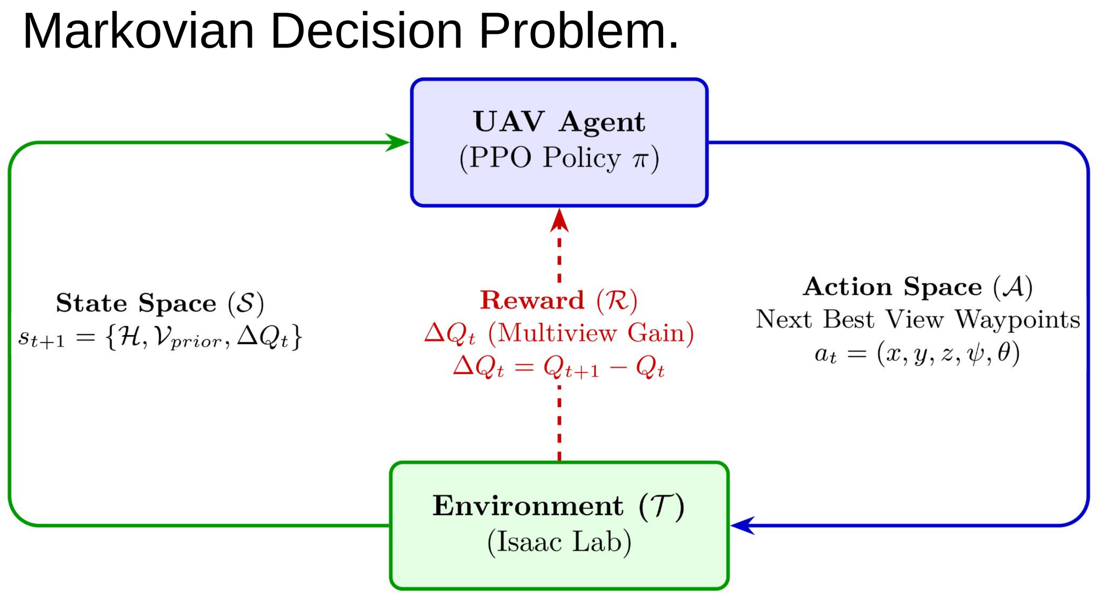
- Critical infrastructure requires regular inspection; manual surveys are costly and hazardous [1].
- Predefined patterns (spirals, lawnmowers) assume uniform geometry, they miss occluded surfaces and fail to generalize [2].
- No prior work validates a complete learning-based viewpoint selection pipeline on a fully autonomous drone [3]

## MOTIVATION

RL discovers how to see structures the way an expert inspector does. Being able to identify uncovered surfaces, consider photogrammetry parameters and adapt to any geometry without hand-crafted rules.

## MDP FORMULATION

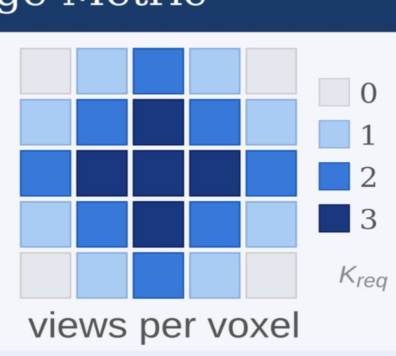
We formulate the viewplanning problem as an Markovian Decision Problem.



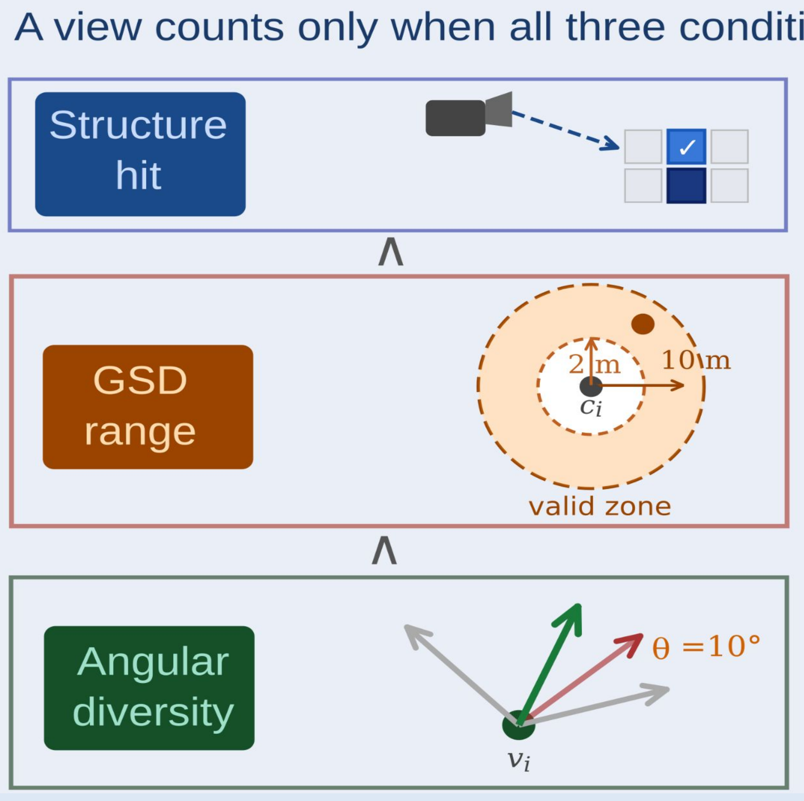

The goal is to learn an optimal policy that maximizes expected cimulative reward.

### Multi-View Coverage Metric

$$Q_t = \frac{\sum_{v \in S} \min(\nu_t(v), K_{req})}{|S| \cdot K_{req}}$$

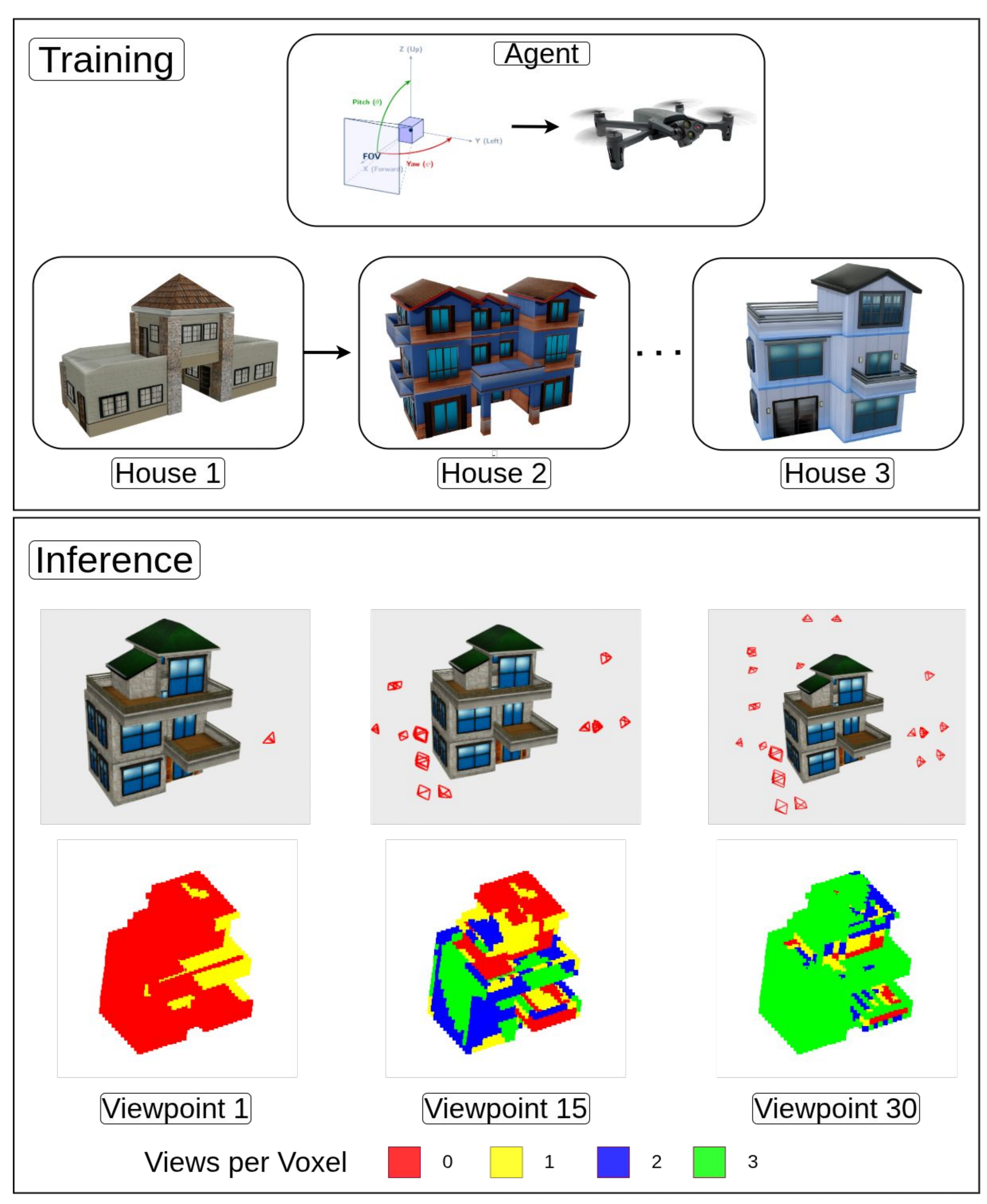



A view counts only when all three conditions hold:

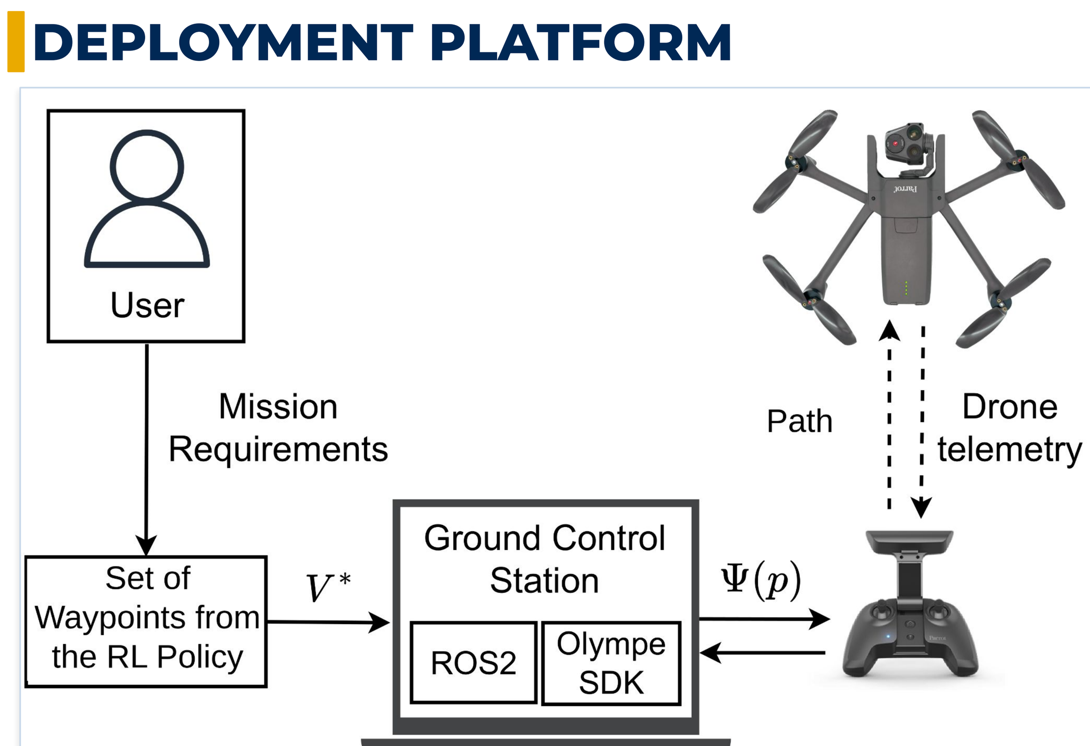



## OBJECTIVES

- **Develop an RL-based View Planning Policy:** Train a RL policy in Isaac Lab for multi-view structural inspection.
- **Optimize for Photogrammetric Constraints:** Integrate GSD, overlap, and incidence angles.
- **Validate Sim-to-Real Transfer:** Deploy the trained policy on a physical UAV.




## DEPLOYMENT PLATFORM

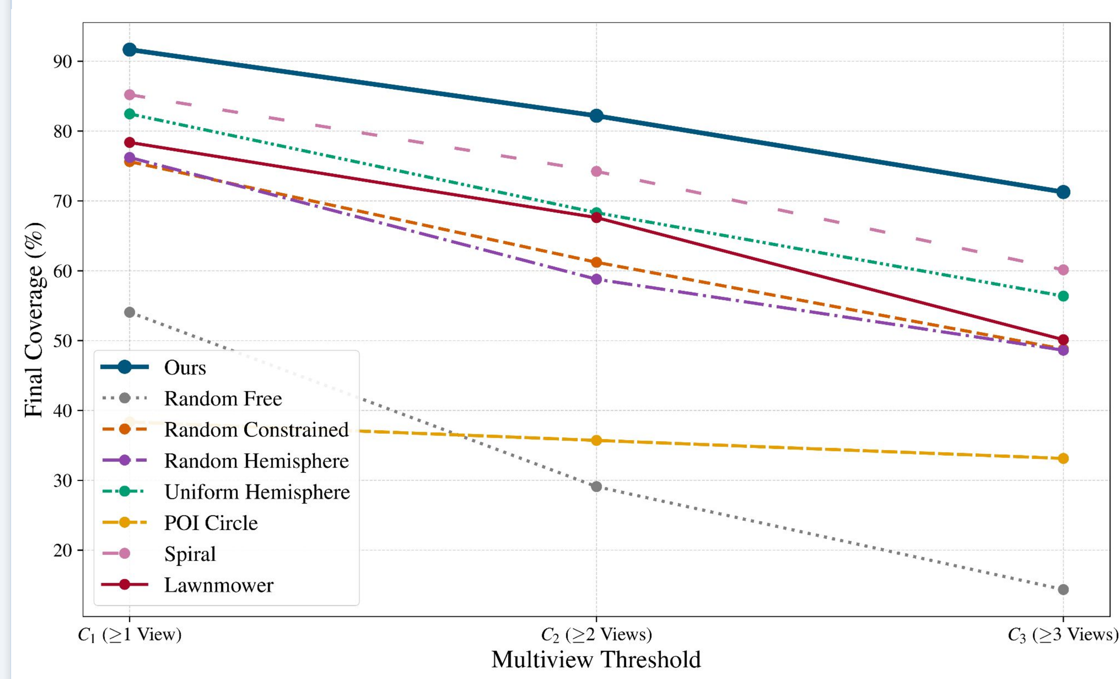



## KEY RESULTS FOR A REAL STRUCTURE

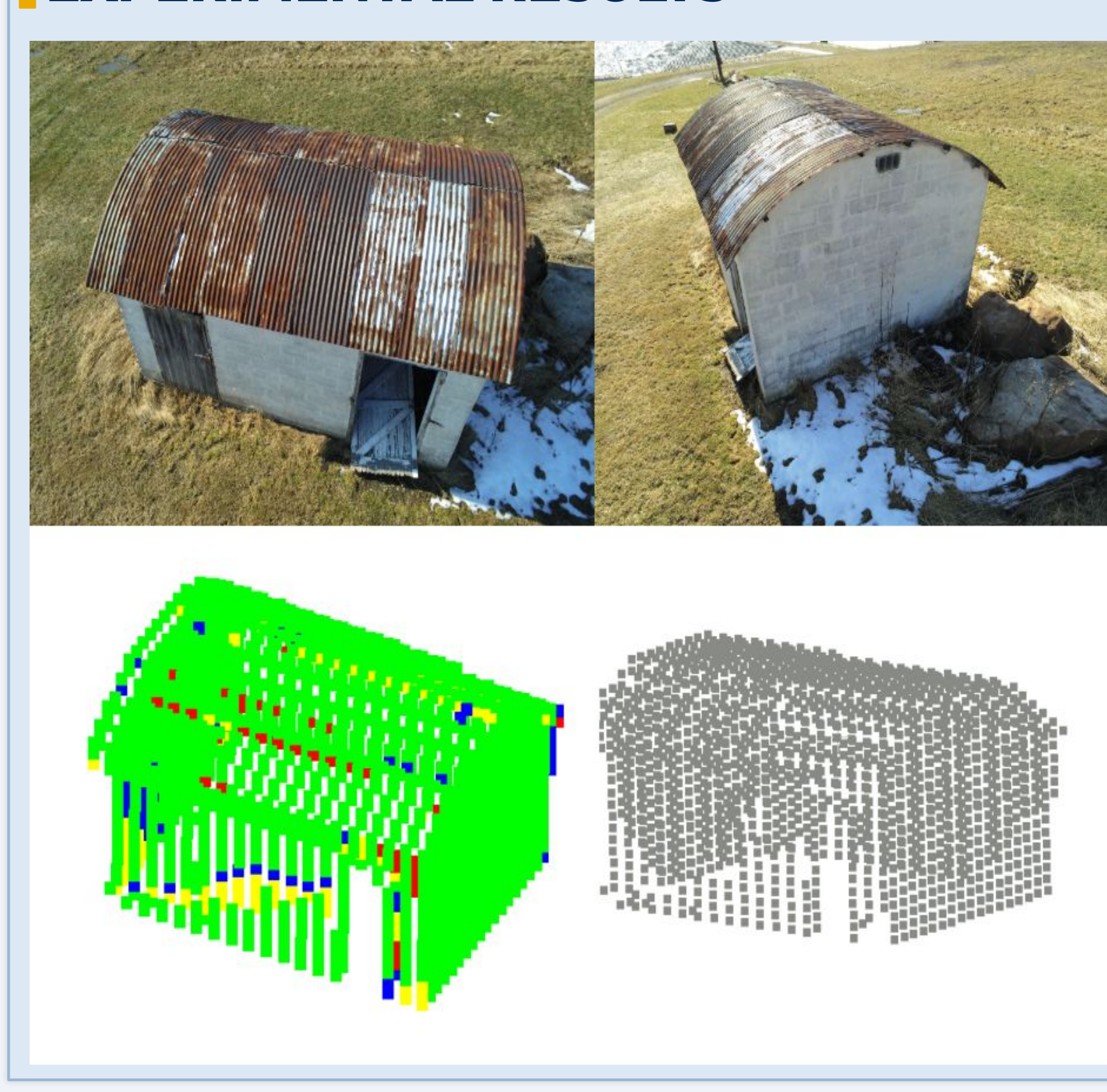
| 96.16% | 89.10% | 33.92% |
|---|---|---|
| C1 Coverage (30 Waypoints) | C2 Triple-Redundancy | Shorter Path Over Best Baseline |
| Best across all methods | Guaranteed inspection reconstruction | vs Spiral / Lawnmower |

## SIMULATED RESULTS

| Method | C1 ↑ | C2 ↑ | C3 ↑ | Path (m) | η (%/m) |
|---|---|---|---|---|---|
| Random Sampling (Free) | 54.06% | 29.10% | 14.36% | 198.19 | 0.16 |
| Random Sampling (GSD) | 75.63% | 61.22% | 48.79% | 127.51 | 0.49 |
| Random Hemisphere | 76.22% | 58.80% | 48.63% | 136.57 | 0.45 |
| Uniform Hemisphere | 82.46% | 68.32% | 56.37% | 161.23 | 0.43 |
| POI Circle | 38.35% | 35.72% | 33.14% | 60.64 | 0.59 |
| Spiral | 85.21% | 74.23% | 60.14% | 243.09 | 0.30 |
| Lawnmower | 78.38% | 67.62% | 50.13% | 121.84 | 0.54 |
| **Ours ★** | **91.66%** | **82.19%** | **71.27%** | **97.03** | **0.84** |




## EXPERIMENTAL RESULTS



## GENERALIZATION ANALYSIS

Our trained policy generalizes to different structures with different geometries withoud the need to adjust parameters or change the architecture.

By limiting the number of waypoints for all methods and comparing them, our policy highest efficiency across all 8 methods on different structures.

## CONCLUSIONS

We developed the first complete RL-based inspection pipeline deployed on a fully autonomous drone across different structures, validated in the real world with zero tuning from across inspections.

Superior coverage across all structure tested was achieved through sim-to-real transfer alone with the highest efficiency (coverage per area) among the baselines compared.

## REFERENCES & FUNDING

[1] H. Shakhatreh et al., "UAVs: A Survey on Civil Applications," IEEE Access, vol. 7, pp. 48572-48634, 2019.
[2] E. Galceran and M. Carreras, "A survey on coverage path planning for robotics," Robot. Auton. Syst., vol. 61, no. 12, pp. 1258-1276, 2013.
[3] A. Bircher et al., "Receding horizon path planning for 3D exploration and surface inspection," Auton. Robots, vol. 42, pp. 291-306, 2018.

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West Virginia University
BENJAMIN M. STATLER COLLEGE OF ENGINEERING AND MINERAL RESOURCES